

# Polishing SOP

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## Step 1

Go to the Project Support Center and check out a diamond-edged saw blade. They should give you a bag with a couple blades in a cardboard container, a couple wrenches, and some metal pieces. Go back to the polishing lab to the saw machine. Lift up the platform to connect the saw blade to the machine. Locate the bolt directly under and to the right of the slit on the platform where the saw blade goes. Put the widest ring on the bolt, then the blade, then the other wide ring. The flat side of the wide rings should be touching the blade. Find the piece that fits into the divet of the wide ring, then screw the nut into the end of the bolt to hold everything in place. Tighten using the two wrenches. Make sure the water valve in the back is closed, should be parallel to the back of the machine. Fill up the water bucket with water from the fume hood and fill up the machine until about 1/4-1/2" of the blade is sitting in the water. Lower the table back down. Place sample against the two stabilizing rods, then clamp in place. Make sure the blade cover is down and the saw is not touching the sample. Turn machine on using the switch on the right side towards the back, then press the green button on the side, then the green button in front. Get RPM's to around 1400. Slowly rotate the handle to move the saw, slowly cutting the sample and keeping RPMs around 1000-1100. Listen to make sure there are no weird noises. A little bit of water should be spraying up. If it is not, then check to make sure the blade is touching the water, since not having water could cause the blade to melt. Cut small piece off of sample with diamond-edged saw, repositioning as necessary. Sample should be about 1cm<sup>3</sup>, just a size where it can fit on an SEM stub and won't spill over the edges of the stub. Disassemble saw, open water valve, dry off the machine and saw pieces. Return bag to PSC.

## Step 2

Turn on Bakelite compressor with switch in the back and open water valve to the left of the machine. Press operate. Configure settings according to the laminated sheet hanging above the machine. Place sample with the face you want to polish facing downwards. Lower the chamber using the arrows on the screen. Fill chamber with 3/4 scoop of Bakelite. Place cap piece on top with metal facing down and green marble facing up and flush with the top of the machine. It should be an airtight fit, so it will take some wiggling around to get in completely. Cover the chamber with the lock bar. Press play. Once complete, remove the lock bar, use the up arrow to push the piece out of the chamber, and pull it off the marble. It should come off pretty easily. Turn off the Bakelite oven.

## Step 3

Turn on sanding belt and open the water valve on top to get water flowing down the belt. Sand the puck by moving it side to side across the belt until the sample is completely exposed and all Bakelite has been removed from its surface. Turn off sander and shut the water valve on top of the sander as well as the big, yellow-handled water valve to the left of the machine.

## Step 4

Initial polisher testing: Turn on the machine and open water valve. Gentle settings (from top down) are 35, 30, 150. Set clock to 30 seconds. Press the "link" button in the top right corner well and "H2O" button simultaneously to turn on the water, then the "link" and "play" buttons simultaneously to start it spinning. If nothing is wrong, press the stop button. Rinse off all parts of the puck, the metal puck holder, metal sand paper ring, and table the sandpaper sits on to get all previous particles off. Press "link" and "up" arrow simultaneously to pull up the puck holder. Wear gloves when handling the sample, do not contaminate the surface. Take off the splash guard ring and metal sand paper holding ring, then place 320-grit sandpaper. Replace the two rings. Place the puck in the metal holder with sample on the sand paper. Turn on the water and start the sander as described above. Sand while continuously checking the surface, moving on to the next grit only when all scratches are the same size. Water should be positioned so the water is washing the sanded-off particles away from the sample. If not, the particles could scratch the sample. Timer can be set for a certain amount of time. Go through grits 320, 400, 600, 800, 1200 (verify these values, they could be off). Between every grit change, use the water to thoroughly clean gloves, the puck, the metal puck holder, metal sand paper ring, and table the sandpaper sits on to get all previous particles off. There is a 1200 plain and 1200 fine, the only difference is texture, would be worth checking with Claude to see if we actually need to do the 1200 fine. Guy helping us said the 1200 plain is usually sufficient. Then use a polishing pad. Polishing pads can be used multiple times if they are used for the exact same alloy, be careful as not following this rule can cause cross-contamination. Peel off the back to expose the sticky side of the pad. Cut small square of electroplated tape to form a tab that can be used to remove the polishing pad by sticking it on the back and wrapping around the side up to the top. Then place on the table. Once pad is in place and puck is clean, use 1micron alumina suspension (Allide high-tech products, de-agglomerated) for the first polishing step. Apply 6-7 generous circles of alumina, focusing on the center and moving outwards since the alumina will slowly move to the outside of the pad as it spins. Make sure puck is completely dry using compressed air from fume hood, and wipe off excess water from the sides and bottom of the puck using a paper towel. **DO NOT WIPE THE SAMPLE WITH THE PAPER TOWEL!** Before starting, get a clean paper towel to place the puck onto after polishing, as well as a deep glass container and squirt bottle full of deionized water. Once applied, move the puck holder over the top of the polishing pad and use link+down to lower it onto the polishing pad. The puck can only touch cleaned surfaces from now on to prevent foreign particles from entering the polishing space and scratching it. Set time to 3 minutes and press start, don't use water on the polishing pad. Regularly apply alumina to the pad to keep well-lubricated. You will be able to see the alumina moving towards the edges of the pad. The alumina is what is doing the polishing at this point. Every couple minutes, remove from puck holder and wash with DI water, letting excess water drain into the deep glass container, checking for any scratches. Repeat until there are no more visible scratches (verify this piece of info). Never set the polished side down. Wash alumina off of polishing pad in fume hood, as well as all of the polishing machine. Dry off pad with compressed air before putting it in the bag. Put the sticky-side cover on the sticky side of the pad, then place in a bag labeled with the polishing material (such as suspended alumina) and the what metal was polished on it. **NOTES:**

All sand paper pads are single use, and if a pad starts to rip while polishing, replace it. The machine has to be reserved, there's a QR code to get to the website.

## **Step 5**

Take the puck, polished side up, to the fume hood in the deep glass container. Spray with ethanol to clean off the surface and dry with compressed air. There may be a haze over the sample, and if there is, follow the rest of this step. Clean out the glass container twice by rinsing it out with ethanol and dumping it into the chemical waste. Now, the glass container and puck are both considered "clean". Place puck faceup in the glass container and submerge puck completely in methanol so polished surface is not exposed to air. Clean out the ultrasonicator with deionized water, then fill with 1/4-1/2 inch of deionized water. Place glass container into the water with a lid on top so no water gets into the glass container or touches the sample. Put lid on top of the ultrasonicator and turn it on. Bring it back to the fume hood. Remove lid carefully so no water drips into methanol. Dry with compressed air. Repeat ultrasonication if haze is still present.

## **Step 6**

Use vibrating polisher (we'll get to this part later).